

## Q1 Finding Your Way Around (20 pts)

```
6:35 PM
rushi@rushi: ~

rushi@rushi:~$
rushi@rushi:~$ pwd
/home/rushi
rushi@rushi:~$ mkdir workspace
rushi@rushi:~$ ls
Cloud-Devops      Documents  Pictures   Public     Templates
Cloud-Devops-Assignments Downloads  practice   scripts    Videos
Desktop           Music      project    snap       workspace
rushi@rushi:~$ mkdir -p ~/workspace/{docs,image,backup}
rushi@rushi:~$ cd workscape
bash: cd: workscape: No such file or directory
rushi@rushi:~$ cd workspace/
rushi@rushi:~/workspace$ ls
backup docs image
rushi@rushi:~/workspace$ ls -la
total 20
drwxrwxr-x  5 rushi rushi 4096 Aug 13 18:32 .
drwxr-x--- 22 rushi rushi 4096 Aug 13 18:30 ..
drwxrwxr-x  2 rushi rushi 4096 Aug 13 18:32 backup
drwxrwxr-x  2 rushi rushi 4096 Aug 13 18:32 docs
drwxrwxr-x  2 rushi rushi 4096 Aug 13 18:32 image
rushi@rushi:~/workspace$ cd ..
rushi@rushi:~$ pwd
/home/rushi
rushi@rushi:~$ ls -la workspace/
total 20
drwxrwxr-x  5 rushi rushi 4096 Aug 13 18:32 .
drwxr-x--- 22 rushi rushi 4096 Aug 13 18:30 ..
drwxrwxr-x  2 rushi rushi 4096 Aug 13 18:32 backup
drwxrwxr-x  2 rushi rushi 4096 Aug 13 18:32 docs
drwxrwxr-x  2 rushi rushi 4096 Aug 13 18:32 image
rushi@rushi:~$
```

## Q2 Sending Output Where You Want (20 pts)

```
rushi@rushi:~/workspace$ pwd
/home/rushi/workspace
rushi@rushi:~/workspace$ echo "Hello Linux" > notes.txt
rushi@rushi:~/workspace$ echo "Learning redirection" >> notes.txt
rushi@rushi:~/workspace$ cat notes.txt
Hello Linux
Learning redirection
```

```
rushi@rushi:~/workspace$ ls /etc/ > etc-list.txt
rushi@rushi:~/workspace$ ls
backup docs etc etc-list.txt image notes.txt
rushi@rushi:~/workspace$ head -n 5 etc-list.txt
adduser.conf
alsa
alternatives
anacrontab
apache2
rushi@rushi:~/workspace$
```

```
rushi@rushi:~/workspace$ ls
backup docs etc etc-list.txt image notes.txt
rushi@rushi:~/workspace$ etc-list.txt wc -l
etc-list.txt: command not found
rushi@rushi:~/workspace$ wc -l etc-list.txt
227 etc-list.txt
rushi@rushi:~/workspace$
```

### Q3 Copy, Rename, and Remove (20 pts)

```
7:15 PM
rushi@rushi: ~
rushi@rushi: ~/workspace
rushi@rushi:~/workspace$ pwd
/home/rushi/workspace
rushi@rushi:~/workspace$ ls
backup docs etc etc-list.txt image notes.txt
rushi@rushi:~/workspace$ cp notes.txt notes-backup.txt
rushi@rushi:~/workspace$ cat notes-backup.txt
Hello Linux
Learning redirection
rushi@rushi:~/workspace$ cp -r ~/workspace/ ~/wrokspace-copy/
rushi@rushi:~/workspace$ ls
backup docs etc etc-list.txt image notes-backup.txt notes.txt
rushi@rushi:~/workspace$ ls ~
Cloud-Devops Documents Pictures Public Templates wrokspace-copy
Cloud-Devops-Assignments Downloads practice scripts Videos
Desktop Music project snap workspace
rushi@rushi:~/workspace$ mv notes-backup.txt notes-old.txt
rushi@rushi:~/workspace$ ls
backup docs etc etc-list.txt image notes-old.txt notes.txt
rushi@rushi:~/workspace$ mv notes-old.txt ~/workspace/backup
rushi@rushi:~/workspace$ ls
backup docs etc etc-list.txt image notes.txt
rushi@rushi:~/workspace$ ls backup/
notes-old.txt
rushi@rushi:~/workspace$ rmdir image/
rushi@rushi:~/workspace$ ls
backup docs etc etc-list.txt notes.txt
rushi@rushi:~/workspace$ rmdir -r ~/wrokspace-copy/
rmdir: invalid option -- 'r'
Try 'rmdir --help' for more information.
rushi@rushi:~/workspace$ rmdir ~/wrokspace-copy/
rmdir: failed to remove '/home/rushi/wrokspace-copy/': Directory not empty
rushi@rushi:~/workspace$ ls ~/wrokspace-copy/
backup docs etc etc-list.txt image notes-backup.txt notes.txt
rushi@rushi:~/workspace$ cd ..
rushi@rushi:~$ pwd
/home/rushi
rushi@rushi:~$ rm -r ~/wrokspace-copy/
rushi@rushi:~$ ls
Cloud-Devops Documents Pictures Public Templates
Cloud-Devops-Assignments Downloads practice scripts Videos
Desktop Music project snap workspace
rushi@rushi:~$
```

## Q. 4 Who Owns What, and Who Can Do What (20 pts)

```
7:24 PM
rushi@rushi: ~/workspace

rushi@rushi:~$ ls
Cloud-Devops      Documents  Pictures   Public     Templates
Cloud-Devops-Assignments Downloads  practice   scripts    Videos
Desktop           Music      project    snap       workspace
rushi@rushi:~$ cd workspace/
rushi@rushi:~/workspace$ ls
backup docs etc etc-list.txt notes.txt
rushi@rushi:~/workspace$ echo "Hello" > script.sh
rushi@rushi:~/workspace$ ls
backup docs etc etc-list.txt notes.txt script.sh
rushi@rushi:~/workspace$ ls -l script.sh
-rw-rw-r-- 1 rushi rushi 6 Aug 13 19:20 script.sh
rushi@rushi:~/workspace$ chmod u+x script.sh
rushi@rushi:~/workspace$ ls -l script.sh
-rwxr--r-- 1 rushi rushi 6 Aug 13 19:20 script.sh
rushi@rushi:~/workspace$ chmod 700 script.sh
rushi@rushi:~/workspace$ ls -l
total 24
drwxrwxr-x 2 rushi rushi 4096 Aug 13 19:11 backup
drwxrwxr-x 2 rushi rushi 4096 Aug 13 18:32 docs
drwxrwxr-x 3 rushi rushi 4096 Aug 13 18:51 etc
-rw-rw-r-- 1 rushi rushi 2167 Aug 13 19:02 etc-list.txt
-rw-rw-r-- 1 rushi rushi 33 Aug 13 18:56 notes.txt
-rwx----- 1 rushi rushi 6 Aug 13 19:20 script.sh
rushi@rushi:~/workspace$ ls -l script.sh
-rwx----- 1 rushi rushi 6 Aug 13 19:20 script.sh
rushi@rushi:~/workspace$
```

## Q5 The Small Project (20 pts)

```
rushi@rushi: ~/works... x    rushi@rushi: ~ x    rushi@rushi: ~ x    rushi@rushi: ~/projects x    v

rushy@rushy:~$ mkdir -p ~/projects/{raws,final}
rushy@rushy:~$ ls
Cloud-Devops      Documents  Pictures  projects  snap      workspace
Cloud-Devops-Assignments Downloads  practice Public    Templates
Desktop          Music     project  scripts  Videos

rushy@rushy:~$ cd projects/
rushy@rushy:~/projects$ ls
final  raws
rushy@rushy:~/projects$ pwd
/home/rushi/projects
rushy@rushy:~/projects$ ls
final  raws
rushy@rushy:~/projects$ touch data1.txt data2.txt data3.txt
rushy@rushy:~/projects$ mv ~/projects/{data1.txt,data2.txt data3.txt} raw
mv: target 'raw': No such file or directory
rushy@rushy:~/projects$ mv ~/projects/{data1.txt,data2.txt data3.txt} raws/
mv: cannot stat '/home/rushi/projects/{data1.txt,data2.txt}': No such file or directory
mv: cannot stat 'data3.txt}': No such file or directory
rushy@rushy:~/projects$ ls
data1.txt  data2.txt  data3.txt  final  raws
rushy@rushy:~/projects$ mv ~/projects/{data1.txt,data2.txt data3.txt} raws/
mv: cannot stat '/home/rushi/projects/{data1.txt,data2.txt}': No such file or directory
mv: cannot stat 'data3.txt}': No such file or directory
rushy@rushy:~/projects$ mv ~/projects/{data1.txt,data2.txt,data3.txt} raws/
rushy@rushy:~/projects$ ls
final  raws
rushy@rushy:~/projects$ ls raws
data1.txt  data2.txt  data3.txt
rushy@rushy:~/projects$ cd raws/
rushy@rushy:~/projects/raws$ echo "report" > data1.txt
```



```
rushi@rushi:~/projects/raws$ echo "report" > data2.txt
rushi@rushi:~/projects/raws$ echo "report" > data3.txt
rushi@rushi:~/projects/raws$ ls
data1.txt  data2.txt  data3.txt
rushi@rushi:~/projects/raws$ cat data2.txt
report
rushi@rushi:~/projects/raws$ cat data3
cat: data3: No such file or directory
rushi@rushi:~/projects/raws$ cat data3.txt
report
rushi@rushi:~/projects/raws$ cat data1.txt data2.txt data3.txt > final/combined.txt
bash: final/combined.txt: No such file or directory
rushi@rushi:~/projects/raws$ cat data1.txt data2.txt data3.txt > ~/project/final/combined.
txt
rushi@rushi:~/projects/raws$ cd ..
rushi@rushi:~/projects$ ls
final  raws
rushi@rushi:~/projects$ ls final/
rushi@rushi:~/projects$ cat data1.txt data2.txt data3.txt > ~/projects/final/combined.txt
cat: data1.txt: No such file or directory
cat: data2.txt: No such file or directory
cat: data3.txt: No such file or directory
rushi@rushi:~/projects$ cd raws/
rushi@rushi:~/projects/raws$ cat data1.txt data2.txt data3.txt > ~/projects/final/combined
.txt
rushi@rushi:~/projects/raws$ cd ..
rushi@rushi:~/projects$ ls final/
combined.txt
rushi@rushi:~/projects$ rm -r raws/
rushi@rushi:~/projects$ ls
final
```